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timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module WB (
    input wire clk,
    input wire rst_n,

    // From MEM stage
    input wire i_RegSrc,           // Register write enable
    input wire [4:0] i_Rd,         // Destination register
    input wire [1:0] i_ResultSrc,  // Result select: 00 = ALU/AMO, 01 = Mem, 10 =
PC+4

    // Possible writeback sources
    input wire [31:0] i_Result,    // ALU or AMO result
    input wire [31:0] i_Wb_Data,   // Memory read data
    input wire [31:0] i_Pc_4,      // PC + 4

    // Outputs to reg_file
    output wire [4:0] o_Rd,         // Destination reg ID
    output wire o_RegSrc,          // Write enable
    output wire [31:0] o_Wb_data   // Final data to be written to register file
);

    // Assign outputs
    assign o_Rd = i_Rd;
    assign o_RegSrc = i_RegSrc;

    // Writeback data mux:
    // ResultSrc:
    // 00 = ALU result (also used for AMO result)
    // 01 = Data memory result (e.g., LW, LB, etc.)
    // 10 = PC+4 (e.g., for JAL, JALR)
    wire [31:0] wb_data_mux;
    assign wb_data_mux = (i_ResultSrc == 2'b00) ? i_Result :
                        (i_ResultSrc == 2'b01) ? i_Wb_Data :
                        (i_ResultSrc == 2'b10) ? i_Pc_4 : 32'd0;

    // Final write-back data gated with reset
    assign o_Wb_data = (rst_n) ? wb_data_mux : 32'd0;

endmodule

```